

08 12 22L 32
 07 16 21L
 06
 05
 02L
 03

Basic registers of the timer

Both Timer 0 and Timer 1 are 16 bits wide. Since the 8051 has an 8-bit architecture, each 16-bit timer is accessed as two separate registers of low byte and high byte. Each timer is discussed separately.

Timer 0 registers

The 16-bit register of Timer 0 is accessed as low byte and high byte. The low byte register is called TL0 (Timer 0 low byte) and the high byte register is referred to as TH0 (Timer 0 high byte). These registers can be accessed like any other register, such as A, B, R0, R1, R2, etc. For example, the instruction "MOV TL0, #4FH" moves the value 4FH into TL0, the low byte of Timer 0. These registers can also be read like any other register. For example, "MOV R5, TH0" saves TH0 (high byte of Timer 0) in R5.

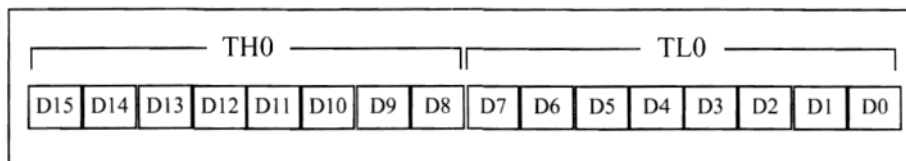


Figure 9-1. Timer 0 Registers

Timer 1 registers

Timer 1 is also 16 bits, and its 16-bit register is split into two bytes, referred to as TL1 (Timer 1 low byte) and TH1 (Timer 1 high byte). These registers are accessible in the same way as the registers of Timer 0.

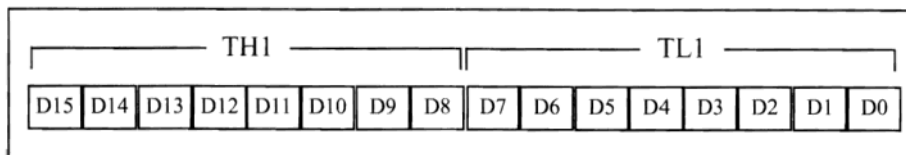


Figure 9-2. Timer 1 Registers

TMOD (timer mode) register

Both timers 0 and 1 use the same register, called TMOD, to set the various timer operation modes. TMOD is an 8-bit register in which the lower 4 bits are set aside for Timer 0 and the upper 4 bits for Timer 1. In each case, the lower 2 bits are used to set the timer mode and the upper 2 bits to specify the operation. These options are discussed next.

(MSB)				(LSB)			
GATE	C/T	M1	M0	GATE	C/T	M1	M0
Timer 1				Timer 0			

GATE Gating control when set. The timer/counter is enabled only while the INTx pin is high and the TRx control pin is set. When cleared, the timer is enabled whenever the TRx control bit is set.

C/T Timer or counter selected cleared for timer operation (input from internal system clock). Set for counter operation (input from Tx input pin).

M1 Mode bit 1

M0 Mode bit 0

M1	M0	Mode	Operating Mode
0	0	0	13-bit timer mode
			8-bit timer/counter THx with TLx as 5-bit prescaler
0	1	1	16-bit timer mode
			16-bit timer/counters THx and TLx are cascaded; there is no prescaler
1	0	2	8-bit auto reload
			8-bit auto reload timer/counter; THx holds a value that is to be reloaded into TLx each time it overflows.
1	1	3	Split timer mode

Figure 9-3. TMOD Register

Example 9-1

Indicate which mode and which timer are selected for each of the following.

- (a) MOV TMOD, #01H (b) MOV TMOD, #20H (c) MOV TMOD, #12H

Example 9-2

Find the timer's clock frequency and its period for various 8051-based systems, with the following crystal frequencies.

- (a) 12 MHz
 (b) 16 MHz
 (c) 11.0592 MHz

Example 9-4

In the following program, we are creating a square wave of 50% duty cycle (with equal portions high and low) on the P1.5 bit. Timer 0 is used to generate the time delay. Analyze the program.

```

                MOV    TMOD,#01        ;Timer 0, mode 1(16-bit mode)
HERE:           MOV    TLO,#0F2H       ;TLO = F2H, the Low byte
                MOV    TH0,#0FFH       ;TH0 = FFH, the High byte
                CPL     P1.5            ;toggle P1.5
                ACALL  DELAY
                SJMP   HERE             ;load TH, TL again
;-----delay using Timer 0
DELAY:
                SETB   TRO              ;start Timer 0
AGAIN:          JNB    TFO,AGAIN        ;monitor Timer 0 flag until
                ;it rolls over
                CLR    TRO              ;stop Timer 0
                CLR    TFO              ;clear Timer 0 flag
                RET
```

Solution: